

Retained diffusion for the actual affine-core periodic profile

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This calculation concerns the precise periodic profile in Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, released 76-page source, Sections 3.1 and 4.1, equations (3.1), (3.2), (4.1)–(4.4). It proves exact identities and a profile evolution for a prescribed affine background, including the exact force that keeps the original profile. It also identifies the change to the affine region when that force is omitted. No conclusion about an infinite viscous controlled sequence follows from these profile identities alone.

1 The original profile and all coefficients

Let $0 < s_0 < 1$ and let $\chi \in C_c^\infty((-\pi, \pi))$ be even and equal to one on $[-s_0, s_0]$. The actual source profile is the 2π -periodic extension of

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad -\pi < s < \pi. \quad (1)$$

It is smooth because it equals $\sin s$ near the endpoints of this period; it is odd because χ is even. Hence its period mean is zero. It equals s on $[-s_0, s_0]$, in particular $F'(0) = 1$. Let P be its unique mean-zero periodic primitive. Indeed $\tilde{P}(s) = \int_0^s F(r) dr$ is even and periodic, and $P = \tilde{P} - \langle \tilde{P} \rangle$ has the claimed properties. Here $\langle h \rangle = (2\pi)^{-1} \int_{-\pi}^{\pi} h(s) ds$.

Fix a closed finite time interval $I = [t_0, t_1]$, smooth $D : I \rightarrow \mathbb{R}^{2 \times 2}$ with $\text{tr } D = 0$, smooth $G : I \rightarrow \mathbb{R}^2$, a frequency $\lambda > 0$, and a nonzero initial vector $\zeta(t_0)$. Define $J(x_1, x_2) = (-x_2, x_1)$ and

$$\dot{\zeta} = -D^T \zeta, \quad s(x, t) = \lambda \zeta(t) \cdot x, \quad \kappa(t) = |\zeta(t)|^2, \quad K(t) = \lambda^2 \kappa(t). \quad (2)$$

The notation $\kappa = |\zeta|^2$ is the source's geometric notation. The physical temperature diffusivity is $\kappa_{\text{th}} \geq 0$, and the physical momentum viscosity is $\nu_{\text{phys}} \geq 0$; neither is the paper's time-scaling parameter named ν . These constants are not set to one. This calculation does not rescale A_0, λ_0, μ , the physical coordinates, time, or laboratory gravity $e_2 = (0, 1)$.

The nonvanishing of ζ follows directly from the invertible fundamental matrix of $\dot{M} = DM$, $M(t_0) = \text{Id}$: differentiation gives $\zeta = M^{-T} \zeta(t_0)$, and differentiation of the determinant gives $\det M = \exp \int_{t_0}^t \text{tr } D = 1$. Write

$$a(t) = -\frac{J\zeta \cdot G}{\lambda \kappa}, \quad b(t) = \lambda \zeta_1. \quad (3)$$

In particular the buoyancy coefficient uses the first laboratory component ζ_1 .

For any smooth scalar θ and divergence-free velocity u , with $\omega = \partial_1 u_2 - \partial_2 u_1$, define the full retained-diffusion residuals by

$$\mathcal{R}_\theta(\theta, u) = \partial_t \theta + u \cdot \nabla \theta - \kappa_{\text{th}} \Delta \theta, \quad \mathcal{R}_\omega(\theta, u) = \partial_t \omega + u \cdot \nabla \omega - \partial_1 \theta - \nu_{\text{phys}} \Delta \omega. \quad (4)$$

Thus these residuals are the required scalar force and curl of the required momentum force. On the region under consideration the older fields are $\theta_{\text{old}} = G \cdot x$, $u_{\text{old}} = Dx$; they need not have zero residual. All statements below subtract their actual residuals.

2 Exact residual and failure of constant-profile damping

Proposition 1. *For arbitrary smooth real amplitudes Θ, Ω , define the source wave*

$$\vartheta = \Theta F(s), \quad \psi = \frac{\Omega}{\lambda^2 \kappa} P(s), \quad v = \nabla^\perp \psi = \frac{\Omega}{\lambda \kappa} J \zeta F(s), \quad \varpi = \Delta \psi = \Omega F'(s). \quad (5)$$

The scalar and curl residual increments are exactly

$$\mathcal{R}_\theta(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\theta(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Theta} - a\Omega)F - \kappa_{\text{th}} K \Theta F'', \quad (6)$$

$$\mathcal{R}_\omega(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\omega(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Omega} - b\Theta)F' - \nu_{\text{phys}} K \Omega F'''. \quad (7)$$

Every profile in these formulas is evaluated at $s(x, t)$.

Proof. Equation (2) gives $(\partial_t + D x \cdot \nabla)s = \lambda(\dot{\zeta} + D^T \zeta) \cdot x = 0$. The orthogonality $J \zeta \cdot \zeta = 0$ gives both $v \cdot \nabla \vartheta = 0$ and $v \cdot \nabla \varpi = 0$. The older vorticity $D_{21} - D_{12}$ is spatially constant, so $v \cdot \nabla \omega_{\text{old}} = 0$. Moreover $v \cdot G = -a\Omega F$, $\partial_1 \vartheta = b\Theta F'$, $\Delta \vartheta = K \Theta F''$, and $\Delta \varpi = K \Omega F'''$. Expanding (4), canceling the older residual, and retaining all these terms proves (6)–(7). The displayed velocity and vorticity in (5) follow by differentiating $P' = F$ twice; their signs follow from the specified J . $\square$

The profile derivatives, including their transition-region terms, are

$$F' = \cos s + \chi'(s)(s - \sin s) + \chi(s)(1 - \cos s), \quad (8)$$

$$F'' = (\chi - 1) \sin s + 2\chi'(1 - \cos s) + \chi''(s - \sin s), \quad (9)$$

$$F''' = (\chi - 1) \cos s + 3\chi' \sin s + 3\chi''(1 - \cos s) + \chi'''(s - \sin s). \quad (10)$$

These follow by the product rule on $(-\pi, \pi)$ and extend smoothly across the period endpoints.

Proposition 2. *There is no constant $c \in \mathbb{R}$ such that $F'' = cF$ on the circle, and no constant $c \in \mathbb{R}$ such that $F''' = cF'$ on the circle. Consequently, at a time with $\kappa_{\text{th}} > 0$ and $\Theta \neq 0$, no choice of scalar $\dot{\Theta} - a\Omega$ makes (6) vanish for every phase. At a time with $\nu_{\text{phys}} > 0$ and $\Omega \neq 0$, no choice of scalar $\dot{\Omega} - b\Theta$ makes (7) vanish for every phase.*

Proof. On $(-s_0, s_0)$, $F(s) = s$ and $F'' = 0$. An identity $F'' = cF$ therefore implies $cs = 0$ for every such s , so $c = 0$. Then $F'' = 0$ everywhere would make F globally affine. Its periodicity would force its slope to be zero, contradicting $F'(0) = 1$. If $F''' = cF'$, evaluation inside that same interval gives $0 = c$, since $F' = 1$ there. Then $F''' = 0$ everywhere makes F a quadratic polynomial on $\mathbb{R}$. Periodicity forces both its quadratic and linear coefficients to vanish, again contradicting $F'(0) = 1$. Since $K > 0$, each asserted residual cancellation with a nonzero amplitude would imply precisely one of these excluded constant-coefficient identities, by division by $\kappa_{\text{th}} K \Theta$ or $\nu_{\text{phys}} K \Omega$. $\square$

For example a proposed common damping equation $\dot{\Theta} = a\Omega - d\Theta$, $\dot{\Omega} = b\Theta - d\Omega$ gives $-d\Theta F - \kappa_{\text{th}} K \Theta F''$ and $-d\Omega F' - \nu_{\text{phys}} K \Omega F'''$ as its two residuals. On the exact affine core they are $-d\Theta s$ and $-d\Omega$. This proves failure of that fixed-profile amplitude prescription, not failure of a viscous controlled construction.

3 An exact evolving-profile map for all diffusivities

We now construct the profile evolution instead of replacing the missing profile dynamics by scalar damping. Let $T(s, t), V(s, t)$ be smooth, mean-zero periodic functions, and let $Q_s = V$, $\langle Q \rangle = 0$. Set

$$\vartheta = T(s, t), \quad \psi = \frac{Q(s, t)}{\lambda^2 \kappa}, \quad v = \frac{J \zeta}{\lambda \kappa} V(s, t), \quad \varpi = V_s(s, t). \quad (11)$$

The time derivative T_t here and below holds phase s fixed.

Proposition 3. *The residual increments for (11) are*

$$T_t - aV - \kappa_{\text{th}}KT_{ss}, \quad \partial_s(V_t - bT - \nu_{\text{phys}}KV_{ss}). \quad (12)$$

Their vanishing is equivalent in the mean-zero class to

$$T_t = aV + \kappa_{\text{th}}KT_{ss}, \quad V_t = bT + \nu_{\text{phys}}KV_{ss}. \quad (13)$$

For the initial data $T(s, t_0) = \Theta_0 F(s)$, $V(s, t_0) = \Omega_0 F(s)$, this system has a unique smooth periodic solution on I given by the following exact Fourier evolution. Write

$$F(s) = \sum_{n \geq 1} f_n \sin(ns), \quad f_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(s) \sin(ns) ds. \quad (14)$$

For each $n \geq 1$, let $U_n(t, t_0)$ solve

$$\partial_t U_n = \begin{pmatrix} -\kappa_{\text{th}}Kn^2 & a \\ b & -\nu_{\text{phys}}Kn^2 \end{pmatrix} U_n, \quad U_n(t_0, t_0) = \text{Id}. \quad (15)$$

Then

$$\begin{pmatrix} T(s, t) \\ V(s, t) \end{pmatrix} = \sum_{n \geq 1} f_n U_n(t, t_0) \begin{pmatrix} \Theta_0 \\ \Omega_0 \end{pmatrix} \sin(ns). \quad (16)$$

Proof. The phase and tangency computations in Proposition 1 remain valid, including $v \cdot \nabla T = v \cdot \nabla V_s = 0$. Now $v \cdot G = -aV$ and $\partial_1 T = bT_s$, which proves (12). The expression inside its second derivative has zero period mean. If its phase derivative vanishes, it is constant in s , and its zero mean forces this constant to be zero. This proves the exact equivalence to (13).

For completeness, each matrix solution in (15) can be defined by the convergent iterated-integral series $U_n = \text{Id} + \sum_{j \geq 1} \int_{t_0 < r_j < \dots < r_1 < t} A_n(r_1) \cdots A_n(r_j) dr_j \cdots dr_1$, where A_n is the displayed matrix. For a fixed n , its j th term is bounded by $(|I| \sup_I \|A_n\|)^j / j!$, so termwise differentiation gives the ODE. Uniqueness follows by integrating the difference and applying the iterated integral estimate, which tends to zero as $j \rightarrow \infty$. For its column solution $y = (y_1, y_2)$, direct differentiation gives

$$\frac{1}{2} \partial_t |y|^2 = -Kn^2(\kappa_{\text{th}}y_1^2 + \nu_{\text{phys}}y_2^2) + (a+b)y_1y_2 \leq \frac{1}{2}(|a| + |b|)|y|^2.$$

Multiplication by $\exp(-\int_{t_0}^t (|a| + |b|))$ and differentiation therefore proves the bound, uniform in n ,

$$|y(t)| \leq e^{\frac{1}{2} \int_{t_0}^t (|a| + |b|)} |y(t_0)|.$$

Repeated differentiation of (15) gives, for each fixed j , $\sup_I |\partial_t^j y| \leq C_j(1+n)^{2j}|y(t_0)|$, by induction: the coefficient derivatives of A_n are bounded by constants times $(1+n)^2$, and the differentiated product has finitely many terms.

Integration by parts any prescribed number of times in (14) shows $|f_n| \leq C_N n^{-N}$ for every N , without endpoint terms by periodicity. Thus (16) and every mixed derivative converge uniformly on $\mathbb{T} \times I$, and termwise differentiation proves the equations and the prescribed initial values. The Fourier expansion indeed equals F : its absolutely convergent series has the same Fourier coefficients as F , and a continuous function with all Fourier coefficients zero is zero. To see this last assertion directly, convolve with the Fejér kernel $\mathfrak{F}_N(r) = N^{-1}(\sin(Nr/2)/\sin(r/2))^2$. It has integral 2π , is nonnegative, and its integral outside every fixed neighborhood of 0 tends to zero because its numerator is at most one and the denominator is bounded below there. Uniform continuity therefore gives uniform convergence of its convolution to the continuous function. When all Fourier coefficients are zero each such convolution is zero, proving the assertion. The same Fourier coefficient ODE and its uniqueness show uniqueness among smooth solutions of (13). $\square$

Proposition 4. *The map in Proposition 3 also realizes the full momentum equation. Let $S_s = T$, $\langle S \rangle = 0$, and retain $Q_s = V$, $\langle Q \rangle = 0$. For a solution of (13), the exact pressure increment is*

$$p^{\text{wave}}(x, t) = -\frac{2\zeta \cdot DJ\zeta}{\lambda^2 \kappa^2} Q(s, t) + \frac{\zeta_2}{\lambda \kappa} S(s, t). \quad (17)$$

With this pressure increment, adding (T, v) to any affine older state preserves its actual vector momentum residual, including retained viscosity ν_{phys} and the unchanged laboratory buoyancy θe_2 .

Proof. Put $c = J\zeta/(\lambda\kappa)$, a vector in this proof. The vector momentum residual increment before adding pressure is

$$(\dot{c} + Dc)V + c(V_t - \nu_{\text{phys}}KV_{ss}) - Te_2.$$

Indeed the transported phase has zero material derivative under Dx , $v \cdot \nabla v = 0$, $v \cdot \nabla(Dx) = Dv$, and $\Delta v = KcV_{ss}$. For every trace-free 2×2 matrix, $DJ = -JD^T$; multiplying its four entries verifies the identity. It gives $\dot{c} = Dc - c\dot{\kappa}/\kappa$ and $\dot{\kappa} = -2\zeta \cdot D\zeta$. The identity $(J\zeta) \cdot D(J\zeta) + \zeta \cdot D\zeta = \kappa \text{tr} D = 0$ shows that $\dot{c} + Dc$ is orthogonal to $J\zeta$. Its component along ζ is therefore

$$\dot{c} + Dc = \frac{2\zeta \cdot DJ\zeta}{\lambda \kappa^2} \zeta.$$

Also $bc - e_2 = -\zeta_2 \zeta / \kappa$, by the two explicit components of $J\zeta = (-\zeta_2, \zeta_1)$. Substituting the second profile equation reduces the residual to

$$\zeta \left[\frac{2\zeta \cdot DJ\zeta}{\lambda \kappa^2} V - \frac{\zeta_2}{\kappa} T \right].$$

The spatial gradient of (17) is its negative, because $\nabla s = \lambda\zeta$, $Q_s = V$ and $S_s = T$. This proves the full momentum statement. Mean-zero periodic primitives exist because the profiles have zero mean, by the same integration argument used for P above. $\square$

Proposition 5. *If $\kappa_{\text{th}} = \nu_{\text{phys}} = \delta \geq 0$, put*

$$\tau(t) = \delta \lambda^2 \int_{t_0}^t |\zeta(r)|^2 dr, \quad F_\tau(s) = \mathcal{H}_\tau F(s) = \sum_{n \geq 1} f_n e^{-n^2 \tau} \sin(ns). \quad (18)$$

Let $\dot{\Theta} = a\Omega$, $\dot{\Omega} = b\Theta$ with the given initial amplitudes. The exact solution in Proposition 3 is

$$T = \Theta F_\tau, \quad V = \Omega F_\tau, \quad Q = \Omega \mathcal{H}_\tau P, \quad \varpi = \Omega \partial_s F_\tau. \quad (19)$$

In particular this is an exact retained-diffusion map with the same prescribed D, G, ζ , all original frequency factors, and laboratory gravity. Each nonzero temperature mode has exact gain

$$\frac{T_n(t)}{T_n(t_0)} = \frac{\Theta(t)}{\Theta_0} e^{-n^2 \tau(t)} \quad (f_n \Theta_0 \neq 0). \quad (20)$$

For vorticity the corresponding ratio is $(\Omega(t)/\Omega_0)e^{-n^2 \tau(t)}$ when $nf_n \Omega_0 \neq 0$.

Proof. The matrix in (15) is $B(t) - \delta K(t)n^2 \text{Id}$, where $B = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$. If U solves $\dot{U} = BU$, differentiation of $e^{-n^2 \tau(t)} U(t, t_0)$ gives exactly (15), including its initial identity. Hence uniqueness proves this factorization and (19). Since derivatives commute termwise with the Fourier series, $(\mathcal{H}_\tau P)_s = \mathcal{H}_\tau F$ and its mean remains zero. This verifies the streamfunction as well as the scalar and vorticity profiles. Formula (20) is division of the actual mode coefficients, retaining its stated nonzero denominator. $\square$

The exact temperature phase norm is

$$\|T(\cdot, t)\|_{L^2(-\pi, \pi)}^2 = \pi \Theta(t)^2 \sum_{n \geq 1} f_n^2 e^{-2n^2 \tau(t)}. \quad (21)$$

This follows by integrating the uniformly convergent squared series and using the exact orthogonality identity

$$\int_{-\pi}^{\pi} \sin(ns) \sin(ms) ds = \pi \mathbf{1}_{n=m}.$$

Thus $\|F_\tau\|_2 \leq e^{-\tau} \|F\|_2$, and for any index with $f_n \neq 0$ one has $\|F_\tau\|_2 \geq \sqrt{\pi} |f_n| e^{-n^2 \tau}$. These are inequalities about phase profiles; an unlocalized affine wave need not be integrable over $\mathbb{R}^2$.

4 Positive heat time and the affine core

Proposition 6. *For every $\tau > 0$, $\mathcal{H}_\tau F$ has no affine restriction to any nonempty open interval of $\mathbb{R}$. In particular it cannot retain the source's nontrivial exact linear phase interval. Also F has infinitely many nonzero Fourier modes.*

Proof. For any $R > 0$ and any nonnegative integer m , the series of m th complex derivatives of $\sum_{n \geq 1} f_n e^{-n^2 \tau} \sin(nz)$ is bounded on $|\operatorname{Im} z| \leq R$ by $\sum_{n \geq 1} |f_n| n^m e^{-n^2 \tau + nR} < \infty$. For instance boundedness of f_n and completion of the square make the tail bounded by a polynomial times $e^{-\tau n^2/2}$. The sum therefore defines an entire function $h(z)$ and has its termwise derivatives. If $h(s) = cs + d$ on a nonempty real open interval, the entire function $h(z) - cz - d$ vanishes identically. The identity principle used here follows from Taylor expansion: a nonzero holomorphic function has a first nonzero Taylor coefficient at each zero where its Taylor series is not identically zero, hence that zero is isolated; an accumulating set of zeros forces the Taylor series to vanish locally, and this local vanishing propagates through the connected domain. Periodicity now gives $2\pi c = h(z + 2\pi) - h(z) = 0$. Oddness gives $d = 0$. Thus every Fourier coefficient $f_n e^{-n^2 \tau}$ would vanish, implying $f_n = 0$ for all n . The Fourier uniqueness argument in Proposition 3 would imply $F = 0$, contradicting $F'(0) = 1$. If F had only finitely many nonzero modes, the same entire-function argument, applied to its finite trigonometric sum at heat time zero, would contradict its interval $F(s) = s$. $\square$

There is nevertheless an exact estimate of the defect deep inside the original interval. Define, for $\tau > 0$ and $d > 0$,

$$\mathcal{E}(d, \tau) = \frac{2}{\sqrt{\pi}} \int_{d/(2\sqrt{\tau})}^{\infty} e^{-r^2} dr \leq \frac{2\sqrt{\tau}}{d\sqrt{\pi}} e^{-d^2/(4\tau)}. \quad (22)$$

The inequality follows from $r \geq d/(2\sqrt{\tau})$ and $\int_a^\infty r e^{-r^2} dr = e^{-a^2}/2$.

Proposition 7. *For $0 < r_0 < s_0$, $d = s_0 - r_0$, and $|s| \leq r_0$,*

$$|\partial_s F_\tau(s) - 1| \leq \|F' - 1\|_\infty \mathcal{E}(d, \tau), \quad (23)$$

$$|F_\tau(s) - s| \leq |s| \|F' - 1\|_\infty \mathcal{E}(d, \tau), \quad (24)$$

$$|\partial_s^m F_\tau(s)| \leq \|F^{(m)}\|_\infty \mathcal{E}(d, \tau) \quad (m \geq 2). \quad (25)$$

Proof. The periodized Gaussian kernel of $\mathcal{H}_\tau$ is $\sum_{k \in \mathbb{Z}} (4\pi\tau)^{-1/2} e^{-(r+2\pi k)^2/(4\tau)}$. To verify this representation, its n th Fourier multiplier is the Gaussian integral $\int_{\mathbb{R}} (4\pi\tau)^{-1/2} e^{-r^2/(4\tau)} e^{-inr} dr = e^{-n^2 \tau}$. One direct verification of this integral differentiates it with respect to n , integrates the Gaussian derivative by parts, and obtains $I'(n) = -2\tau n I(n)$ with $I(0) = 1$; the last normalization is the usual Gaussian integral, obtained by squaring it and integrating in polar coordinates.

Fourier uniqueness then proves the representation. For any periodic bounded function h vanishing on $[-s_0, s_0]$, its periodic lift satisfies $h(s-r) = 0$ whenever $|s| \leq r_0$ and $|r| < d$. The Gaussian representation on the line therefore gives

$$|\mathcal{H}_\tau h(s)| \leq \|h\|_\infty \int_{|r| \geq d} (4\pi\tau)^{-1/2} e^{-r^2/(4\tau)} dr = \|h\|_\infty \mathcal{E}(d, \tau).$$

Apply this to $h = F' - 1$ and $h = F^{(m)}$ for $m \geq 2$, using commutation of derivatives with $\mathcal{H}_\tau$ and $\mathcal{H}_\tau 1 = 1$. Finally $F_\tau(0) = 0$ by oddness; integrating the first bound between 0 and s proves the second bound with the same d . $\square$

The source background in equation (3.3) is obtained because each older layer is exactly affine on the newer support, by equations (4.3)–(4.6). For a heat-evolved older layer with nonzero scalar or velocity amplitude, Proposition 6 removes this exact affine premise. Its gradient at the origin can still be computed, but its values away from the origin are not that gradient times x . The tail estimates give explicit defects rather than an identification of those two fields. Consequently Proposition 5 alone supplies no coupled feedback law for the heat-modified older background.

5 Exact preservation by force and the localized diffusion costs

For the original inviscid amplitude equations $\dot{\Theta} = a\Omega$, $\dot{\Omega} = b\Theta$, Proposition 1 shows that the exact additional scalar and curl forces are

$$f_{\theta, \text{diff}} = -\kappa_{\text{th}} K \Theta F'', \quad E_{\text{diff}} = -\nu_{\text{phys}} K \Omega F'''. \quad (26)$$

They vanish on the entire affine interval $|s| < s_0$. This retains exactly the original core, amplitudes, phase and hence its local affine feedback; it pays a nonzero diffusion force in the rest of the profile. In the unlocalized wave this force is periodic in phase and is not a compactly supported force on $\mathbb{R}^2$.

Here is the exact compact counterpart for the leading fields in source equation (4.2). Let M solve $\dot{M} = DM$, $M(t_0) = \text{Id}$, let $p = \zeta(t_0)$, and let $\ell > 0$. Fix the source's radial $f \in C_c^\infty(B_1)$, $f = 1$ on $B_{1/2}$, and put

$$\begin{aligned} a_{\text{mat}} &= M^{-1}x, & g(a_{\text{mat}}) &= f(a_{\text{mat}}/\ell), \\ \theta^L &= w\Theta F(s)g(a_{\text{mat}}), & \psi^L &= \frac{w\Omega}{\lambda^2\kappa} P(s)g(a_{\text{mat}}), & u^L &= \nabla^\perp \psi^L. \end{aligned} \quad (27)$$

The activation $w(t)$ remains the original smooth scalar function; no time differentiation of it is omitted from the existing inviscid residual. The following formulas give exactly the additional diffusion terms, with w retained. At a fixed time put

$$B = M^{-1}, \quad H = BB^T, \quad q = B\zeta, \quad \mathcal{L} = \lambda^2\kappa\partial_s^2 + 2\lambda q \cdot \nabla_a \partial_s + H : \nabla_a^2. \quad (28)$$

Although $s = \lambda p \cdot a_{\text{mat}}$ upon evaluation, s, a are independent variables in this differential operator.

Proposition 8. *For every smooth profile $A(s, a)$, $\Delta_x[A(\lambda\zeta \cdot x, Bx)] = (\mathcal{L}A)(\lambda\zeta \cdot x, Bx)$. Consequently the exact added forces for (27) are*

$$f_{\theta, \text{diff}}^L = -\kappa_{\text{th}} w \Theta [\lambda^2 \kappa F'' g + 2\lambda F' q \cdot \nabla g + F(H : \nabla^2 g)], \quad (29)$$

$$f_{u, \text{diff}}^L = -\nu_{\text{phys}} \Delta u^L = -\nu_{\text{phys}} \nabla^\perp \Delta \psi^L, \quad E_{\text{diff}}^L = -\nu_{\text{phys}} \frac{w\Omega}{\lambda^2\kappa} \mathcal{L}^2(Pg). \quad (30)$$

Writing $\mathcal{H} = H : \nabla_a^2$, the last operator is explicitly

$$\begin{aligned} \mathcal{L}^2(Pg) &= \lambda^4 \kappa^2 F''' g + 4\lambda^3 \kappa F'' q \cdot \nabla g + 2\lambda^2 \kappa F' \mathcal{H} g + 4\lambda^2 F' (q \cdot \nabla)^2 g \\ &\quad + 4\lambda F(q \cdot \nabla) \mathcal{H} g + P \mathcal{H}^2 g. \end{aligned} \quad (31)$$

All these forces are supported in $M\bar{B}_\ell$. They vanish in the exact affine region $MB_{\ell/2} \cap \{|s| < s_0\}$.

Proof. The chain rule gives $\partial_{x_i} A = \lambda \zeta_i A_s + \sum_j B_{ji} A_{a_j}$. Applying it a second time and summing in i gives the three terms in (28), since $\sum_i B_{ji} \zeta_i = (B\zeta)_j$ and $\sum_i B_{ji} B_{ki} = (BB^T)_{jk}$. Applying the same identity twice to Pg proves (30). Expanding the square of the commuting constant-in- (s, a) operators in (28), and using $P' = F$, $P'' = F'$, $P''' = F''$, $P'''' = F'''$, proves all six terms of (31). Spatial differentiation does not enlarge the support of a smooth compactly supported function, proving the support statement. On the plateau $g = 1$ all positive derivatives of g vanish. On the linear phase interval $F'' = F''' = 0$. Thus the scalar and curl formulas vanish on their intersection. There $\psi^L = (w\Omega/(\lambda^2\kappa))(P(0) + s^2/2)$ is quadratic in x , so u^L is linear and $\Delta u^L = 0$, proving the vector-force statement as well. Finally $\text{curl}(-\nu_{\text{phys}}\Delta u^L) = -\nu_{\text{phys}}\Delta^2\psi^L$, so the specified vector force realizes the curl force without any inverse-curl ambiguity. $\square$

For explicit quantitative costs let $C_j = \sup_a \|\nabla^j f(a)\|_{\text{HS}}$, with the Euclidean Hilbert–Schmidt tensor norm, and $C_0 = \|f\|_{\infty}$. Then $\|\nabla^j g\|_{\infty} = C_j \ell^{-j}$ by the chain rule. Write $h = \|H\|_{\text{HS}}$ and $q_* = |q|$. Cauchy–Schwarz on the tensor contractions in (29) and (31) proves

$$\|f_{\theta, \text{diff}}^L\|_{\infty} \leq \kappa_{\text{th}} |w\Theta| [\lambda^2 \kappa \|F''\|_{\infty} C_0 + 2\lambda q_* \|F'\|_{\infty} C_1 \ell^{-1} + h \|F\|_{\infty} C_2 \ell^{-2}], \quad (32)$$

$$\begin{aligned} \|E_{\text{diff}}^L\|_{\infty} &\leq \frac{\nu_{\text{phys}} |w\Omega|}{\lambda^2 \kappa} [\lambda^4 \kappa^2 \|F''''\|_{\infty} C_0 + 4\lambda^3 \kappa q_* \|F'''\|_{\infty} C_1 \ell^{-1} \\ &\quad + (2\lambda^2 \kappa h + 4\lambda^2 q_*^2) \|F'\|_{\infty} C_2 \ell^{-2} + 4\lambda q_* h \|F\|_{\infty} C_3 \ell^{-3} + h^2 \|P\|_{\infty} C_4 \ell^{-4}]. \end{aligned} \quad (33)$$

The same reasoning applies to every compact correction: for a specified smooth correction (θ^c, ψ^c) the extra scalar and vector forces are exactly $-\kappa_{\text{th}}\Delta\theta^c$ and $-\nu_{\text{phys}}\nabla^{\perp}\Delta\psi^c$, with curl $-\nu_{\text{phys}}\Delta^2\psi^c$. This is the exact linear map from an already constructed inviscid forced field to a retained-diffusion forced field:

$$(\theta, u, p, f_{\theta}, f_u) \mapsto (\theta, u, p, f_{\theta} - \kappa_{\text{th}}\Delta\theta, f_u - \nu_{\text{phys}}\Delta u). \quad (34)$$

Substitution in the scalar and momentum equations proves the map: adding $\kappa_{\text{th}}\Delta\theta$ or $\nu_{\text{phys}}\Delta u$ to the new right-hand side cancels its respective force decrement. Thus pressure, divergence, initial data and the entire trajectory are retained exactly. On each closed finite smooth stage these are smooth finite forces; where the original fields and their derivatives are compactly supported, the new terms have the same support. This statement supplies no uniform bound as the layer index tends to infinity. The explicit factors λ^2 , ℓ^{-2} , and ℓ^{-4} above must be retained in such an analysis.

Source and scope receipt

The primary source read here is the 76-page PDF *Blowup for the Boussinesq equations with smooth forcing*, Levent Alpöge and Tristan Buckmaster; source equations (3.1)–(3.6) are on pages 8–9 and (4.1)–(4.6) on pages 37–38. The exact file SHA-256 is

895a628d1783bcb039374686f50b895b5f450f53b8ef8aa173523487a7a4a21b.

The companion replay script checks the profile differentiation, full two-dimensional residual algebra, localized Laplacian square, common diffusivity matrix factorization, and eigenfunction obstruction identities. Its checks supplement the proofs above; they do not certify an infinite-stage existence or force-smoothness theorem.